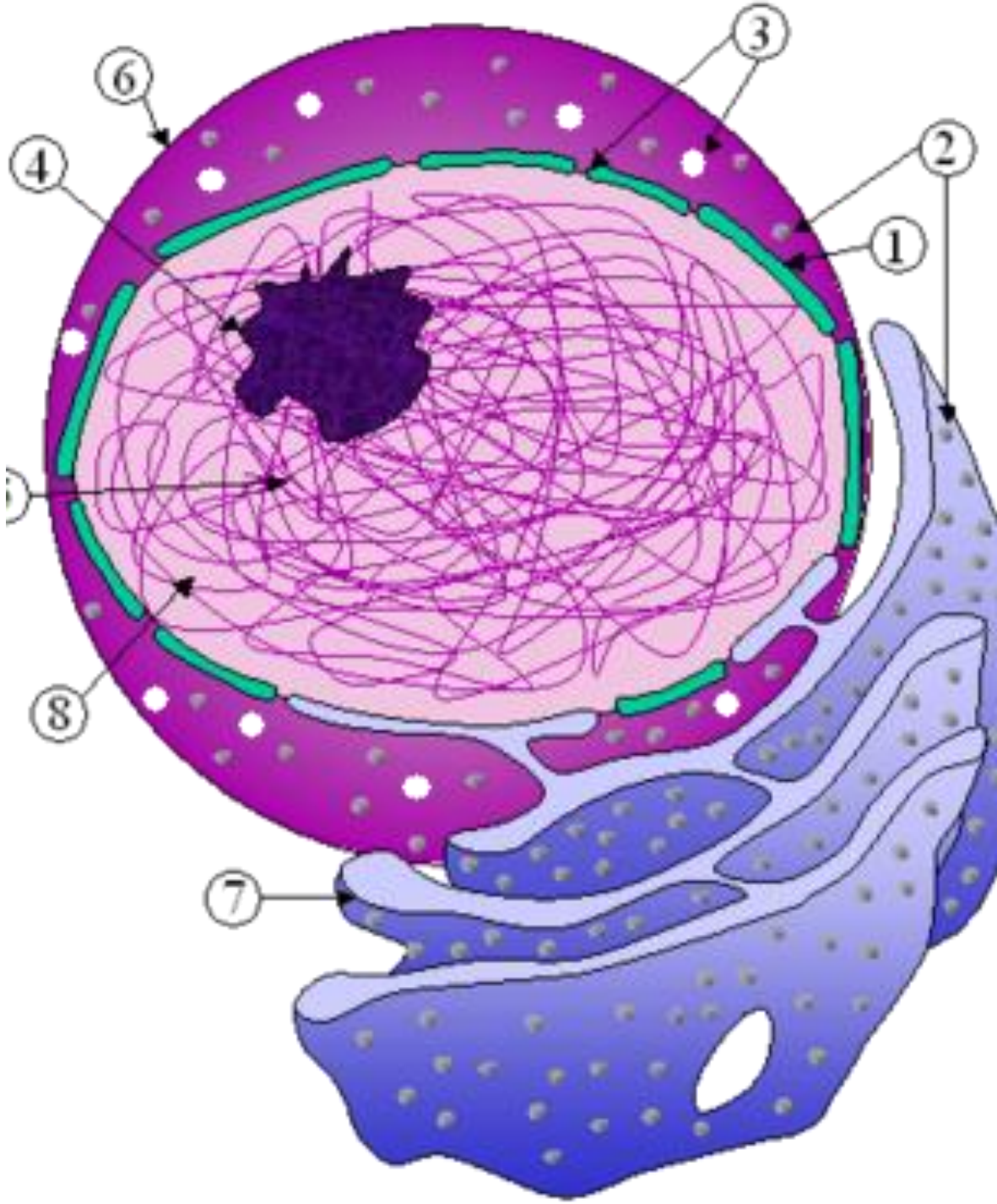
DP Biology – B2.2

Organelles

- How does the structure of each organelle suit its function?
- Why is it useful for cells to have separate compartments?



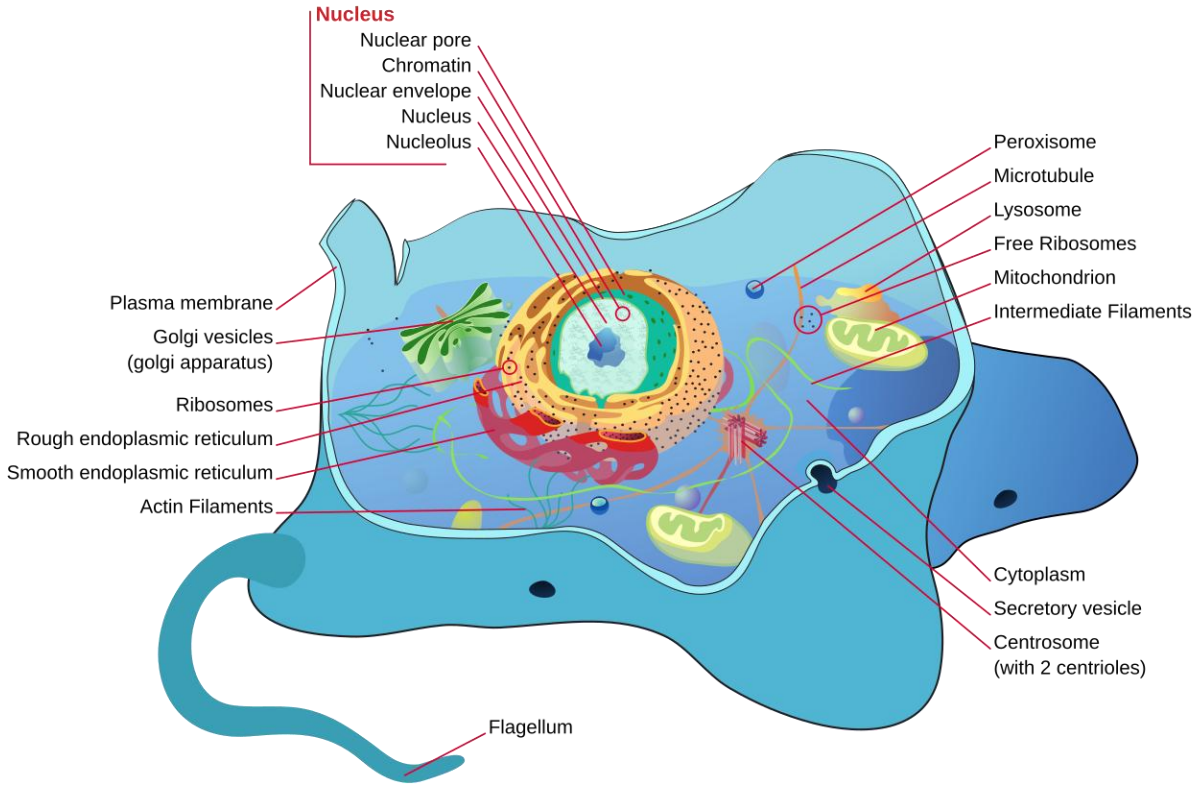
SL statements B2.2.1–3. Textbook pp. 244–247. The structure of each organelle in detail is at HL.
Image: the nucleus and endoplasmic reticulum. Magnus Manske, public domain.

Objectives for each class

Class	Statements	By the end of the class, you can…
1	B2.2.1	Explain what counts as an organelle, and sort cell structures by the number of membranes they have
2	B2.2.2–3	Explain the advantages of separating the nucleus from the cytoplasm, and of compartments in the cytoplasm

What is an organelle?

B2.2.1



An animal cell. LadyofHats, public domain.

- An **organelle** is a discrete subunit of a cell
- Each is adapted to carry out one or more particular functions
- Together they divide up the work of the cell

Key idea

Structure suits function: this is the first guiding question.

Common mistake

Not everything in a cell is an organelle. The cell wall, cytoskeleton and cytoplasm are not. The nucleus, vesicles, ribosomes and plasma membrane are.

Sorting structures by membranes

B2.2.1

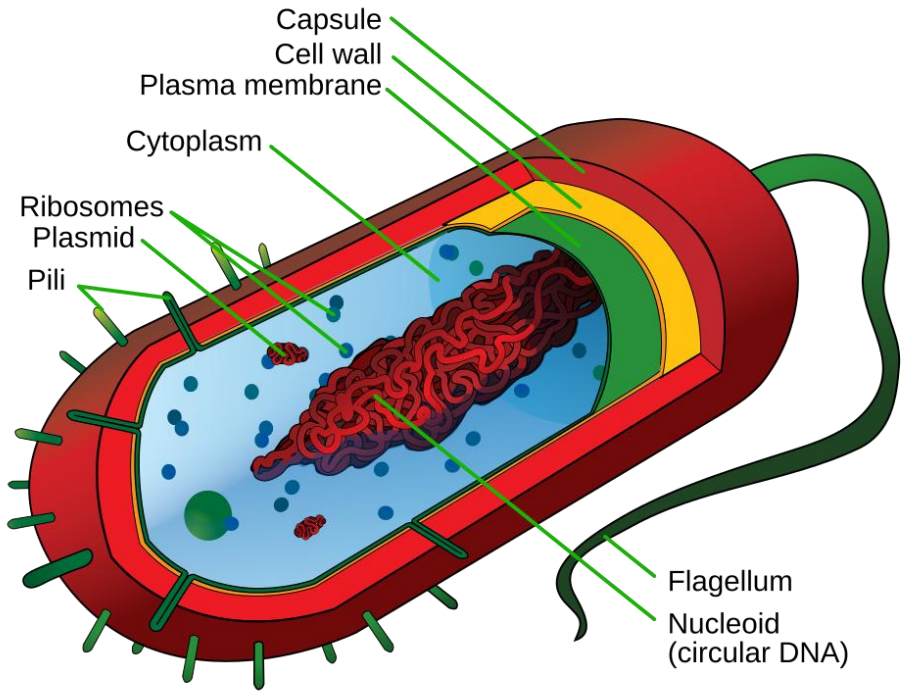
Number of membranes	Examples
Two	Nucleus, mitochondrion, chloroplast
One	Vesicle, vacuole, lysosome, rough and smooth ER, Golgi apparatus
None	Ribosome
Not organelles	Cell wall, cytoskeleton, cytoplasm

Activity

Close the table. Sort these from memory: nucleus, vacuole, ribosome, cytoplasm, chloroplast, cell wall, Golgi apparatus, cytoskeleton, mitochondrion and rough ER.

Prokaryotes have few organelles

B2.2.1



A prokaryotic cell. LadyofHats, public domain.

- Ribosomes and the plasma membrane are found in both prokaryotes and eukaryotes
- Prokaryotes have no membrane-bound organelles
- Eukaryotes are **compartmentalized**: they have separate membrane-bound regions

Practical skill

Biologists separate organelles by spinning broken cells in a centrifuge. Heavier parts settle first, so each fraction can be studied on its own.

Separating the nucleus and cytoplasm

B2.2.2

	Eukaryote	Prokaryote
DNA	Inside a nucleus	In the cytoplasm, as a nucleoid
Transcription and translation	Separate in place and time	Can happen together
mRNA	Modified before it meets ribosomes	Ribosomes may attach straight away

Key idea

The nuclear membrane keeps transcription away from translation. Eukaryotic mRNA can therefore be modified after transcription, before it reaches any ribosome. In prokaryotes this is not possible, because ribosomes can start on the mRNA while it is still being made.

Compartments in the cytoplasm

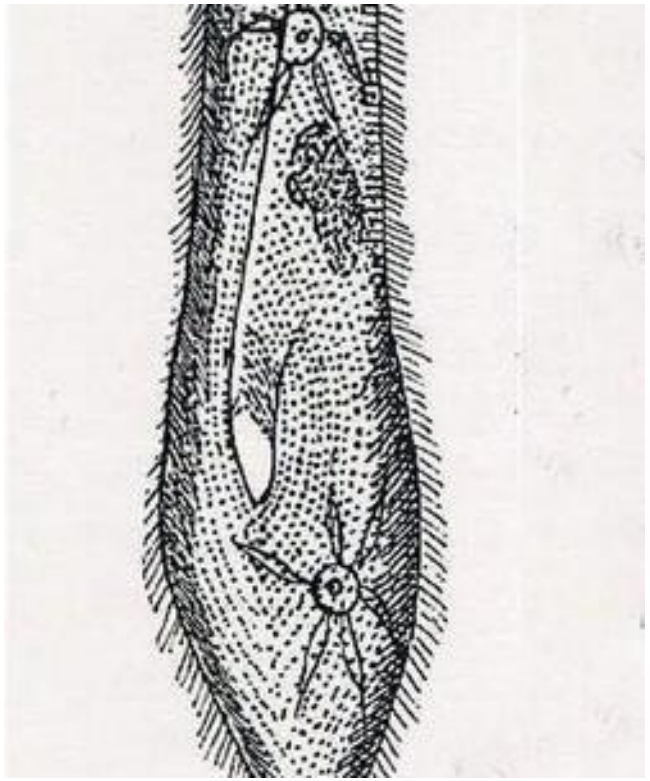
B2.2.3

Advantage	What it means
Enzymes and substrates are concentrated	Reactions happen faster than if they were spread through the cytoplasm
Harmful substances are kept apart	Other parts of the cell are protected from them
Conditions can be different	Each compartment can have its own pH, for example
Incompatible processes are separated	Reactions that would interfere do not meet

Discussion

Use the example of a lysosome: why would releasing its enzymes into the cytoplasm be damaging to the cell?

Two examples: lysosomes and vacuoles



Paramecium. A. Kahl, public domain.

- **Lysosomes** hold digestive enzymes at a high concentration, with an acid pH inside
- The enzymes do not attack the rest of the cell, as they are kept behind a membrane
- **Phagocytic vacuoles** form when a cell engulfs food or bacteria

Example

In Paramecium, food enters a vacuole by endocytosis. Conditions inside change as digestion goes on: first the pH drops, then it rises again.

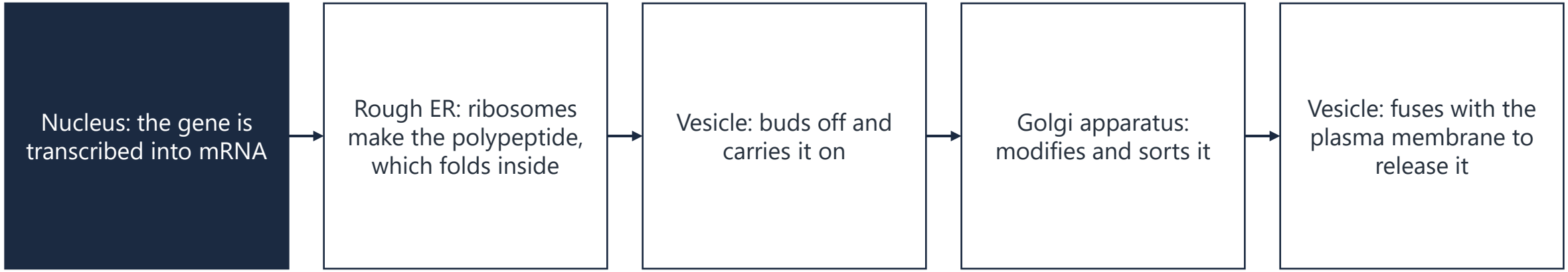
Link

Phagocytes use the same route to take in and digest pathogens (C3.2).

Going further: how compartments work together

B2.2.3

A protein made for export passes through several compartments in turn. Each one does a different job, and vesicles carry the protein between them.



Why this matters

Keeping each step in its own compartment lets each one hold the right enzymes and conditions, and stops new proteins mixing with the rest of the cytoplasm.

Going further: separating organelles

Biologists study organelles by breaking cells open and spinning the mixture in a centrifuge. Heavier parts sink first, so each spin at a higher speed collects smaller organelles.

Spin speed	What settles out
Low	Nuclei
Medium	Mitochondria (and chloroplasts in plant cells)
High	Pieces of ER and other membranes
Very high	Ribosomes

Think

Why must the cells be kept cold and in a solution with the same solute concentration as the cytoplasm?

Check-in: organelles and compartments

B2.2.1–3

1. State what is meant by an organelle.
2. Give two cell structures with no membrane, two with one membrane and two with two membranes.
3. Explain why the cell wall and the cytoskeleton are not usually called organelles.
4. Explain one advantage of separating the nucleus from the cytoplasm.
5. Explain why lysosomes need their own compartment.

How to use this

Answer without your notes, then compare with a partner.

Working with data: enzymes in cell fractions

B2.2.1

Activity of two enzymes in the fractions from a centrifuge (percentage of total). These data are invented for practice.

Fraction	Enzyme X / %	Enzyme Y / %
Nuclei	4	2
Mitochondria	88	6
Membranes	5	81
Liquid left	3	11

1. Identify where enzyme X is found.
2. Suggest an enzyme that enzyme Y might be.
3. Explain why a little activity is found in every fraction.

Common mistakes to avoid

Not this	But this
Ribosomes are surrounded by a membrane.	Ribosomes have no membrane.
Prokaryotes have no organelles.	They have ribosomes, but no membrane-bound organelles.
The nucleus has a single membrane.	The nuclear envelope is a double membrane with pores.

Check that you can…

1. **Define** an organelle and **classify** cell structures by their membranes
2. **Explain** the advantages of separating the nucleus from the cytoplasm
3. **Explain** the advantages of compartments in the cytoplasm, with examples
4. **Outline** how a protein moves through the secretory pathway

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State one structure that is an organelle and one that is not.
2. Explain the advantage of keeping transcription and translation apart.
3. Explain how lysosomes show an advantage of compartmentalization.

Image credits

Nucleus ER.png – Magnus Manske (talk) – Public domain
Animal cell structure en.svg – LadyofHats (Mariana Ruiz) – Public domain
Average prokaryote cell- en.svg – Mariana Ruiz Villarreal, LadyofHats – Public domain
Paramecium caudatum.jpg – Alfred Kahl – Public domain

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